

Introduction to Operational Amplifiers and Op-Amps with Reactive Elements

Rori Bowman

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Abstract

These experiments investigated the behavior of operational amplifier circuits and their response to signals over a range of frequencies. In the first part of the experiment, basic op-amp configurations, including the voltage follower and inverting amplifier, were constructed and analyzed to study gain, bandwidth, and the effects of non-ideal behavior. The frequency response of each circuit was measured using sinusoidal inputs, and Bode plots were generated to determine cutoff frequencies and overall performance. Additionally, the slew rate of the op-amp was measured from the slope of the output waveform during a square-wave transition to evaluate limitations in the rate of change of the output signal.

Then, reactive components were introduced into op-amp circuits to produce frequency-dependent behavior. High-pass and low-pass filters were constructed, and their gain and phase responses were measured across a range of frequencies. The cutoff frequencies were identified from the Bode plots and compared with theoretical predictions.

The results showed that the voltage follower maintained approximately unity gain over a wide frequency range, while the inverting amplifier produced a gain close to the expected value determined by the resistor ratio. The active filters displayed the expected behavior, with cutoff frequencies consistent with theoretical values. Overall, the experiment demonstrated how op-amp circuits can be used to control gain and shape frequency response.

1 Introduction

Operational amplifiers allow for more precise manipulation of electric signals. Through the use of negative feedback, op-amp circuits can be designed to produce stable and defined gain. In practice, however, real op-amps deviate from ideal behavior due to limitations including finite open-loop gain, limited bandwidth, and a finite slew rate. These limitations become especially significant at higher frequencies and large signal amplitudes.

In Lab 8, basic op-amp configurations were studied, including the voltage follower and inverting amplifier. These circuits were used to investigate gain, frequency response, and impedance. The effects of finite open-loop gain and slew rate were examined in order to understand the limitations of real amplifiers and their impact on circuit performance.

In Lab 9, reactive elements were introduced into op-amp circuits to produce frequency-dependent behavior. By incorporating capacitors, the circuits were configured to be high-pass and low-pass filters. The frequency response of these circuits was analyzed and compared to theoretical expectations.

Together these experiments provide a deep understanding of both the ideal and non-ideal behavior of operational amplifiers, as well as their ability to perform signal transformations.

2 Theoretical Framework

These experiments examine the frequency response of operational amplifier circuits and active filters. The response of each circuit is described using the voltage gain,

$$G = \frac{V_{out}}{V_{in}}, \quad (1)$$

where V_{in} and V_{out} are the input and output voltage amplitudes. The gain is often expressed in decibels as

$$G_{dB} = 20 \log_{10} \left| \frac{V_{out}}{V_{in}} \right|. \quad (2)$$

A Bode magnitude plot shows the gain as a function of frequency on a logarithmic scale, making it possible to identify passband behavior, rolloff, and cutoff frequencies. The cutoff frequency is defined as the point where the gain decreases by 3 dB from its maximum value, corresponding to

$$\frac{G}{G_{\max}} = \frac{1}{\sqrt{2}} \approx 0.707. \quad (3)$$

Ideal Op-Amp Behavior

For an ideal operational amplifier operating with negative feedback, the input voltages are approximately equal,

$$v_+ \approx v_-, \quad (4)$$

and the input currents are approximately zero,

$$i_+ \approx i_- \approx 0. \quad (5)$$

These conditions allow the closed-loop behavior of the circuit to be determined primarily by the external components rather than the internal gain of the op-amp.

Slew Rate

A real op-amp cannot change its output voltage immediately. The slew rate (SR) is defined as the maximum rate of change of the output voltage,

$$SR = \max \left| \frac{dV_{out}}{dt} \right|. \quad (6)$$

It can be measured experimentally from the slope of the output during a square-wave

$$SR \approx \left| \frac{\Delta V_{out}}{\Delta t} \right|. \quad (7)$$

For a sinusoidal signal,

$$V_{out}(t) = V_p \sin(2\pi ft). \quad (8)$$

Voltage Follower

In the voltage follower configuration, the output is connected directly to the inverting input. Negative feedback requires

$$v_- \approx v_+ = v_{in}, \quad (9)$$

so the output follows the input,

$$v_o \approx v_{in}. \quad (10)$$

So, the ideal closed-loop gain is

$$G = \frac{v_o}{v_{in}} \approx 1. \quad (11)$$

For a real op-amp with finite open-loop gain $A(f)$,

$$v_o = A(f)(v_{in} - v_o), \quad (12)$$

which leads to

$$G(f) = \frac{A(f)}{A(f) + 1}. \quad (13)$$

At low frequencies the gain is close to unity, while for higher frequencies it decreases as $A(f)$ becomes smaller.

Inverting Amplifier

The inverting amplifier uses an input resistor R_{in} and a feedback resistor R_f , with the non-inverting input grounded. Under negative feedback,

$$v_- \approx 0. \quad (14)$$

The input current is

$$i_{in} = \frac{v_i}{R_{in}}, \quad (15)$$

and this current flows through the feedback resistor,

$$i_f = \frac{-v_o}{R_f}. \quad (16)$$

Setting these equal gives

$$\frac{v_i}{R_{in}} = \frac{-v_o}{R_f}, \quad (17)$$

so the gain is

$$G = \frac{v_o}{v_i} = -\frac{R_f}{R_{in}}. \quad (18)$$

At higher frequencies, the finite open-loop gain reduces the measured gain. The gain-bandwidth product remains approximately constant,

$$\text{GBP} \approx G_{\text{mid}} f_{3dB}. \quad (19)$$

Capacitive Reactance and RC Behavior

The frequency dependence of capacitors is described by the capacitive reactance,

$$X_C = \frac{1}{2\pi f C}. \quad (20)$$

At low frequencies, X_C is large and the capacitor behaves like an open circuit, while at high frequencies it behaves like a short circuit. This allows RC combinations to selectively pass or attenuate different frequency ranges. The characteristic cutoff frequency is

$$f_c = \frac{1}{2\pi RC}. \quad (21)$$

Integrator and Differentiator

In an op-amp integrator, the gain is frequency dependent and given by

$$G(\omega) = -\frac{1}{j\omega R_{in} C_f}, \quad (22)$$

which corresponds in the time domain to

$$v_o(t) = -\frac{1}{R_{in} C_f} \int v_{in}(t) dt. \quad (23)$$

The magnitude of the gain decreases with frequency, producing a slope of approximately -20 dB per decade over the operating range.

For the differentiator, the gain is

$$G(\omega) = -j\omega R_f C_{in}, \quad (24)$$

which corresponds to

$$v_o(t) = -R_f C_{in} \frac{dv_{in}}{dt}. \quad (25)$$

The gain increases with frequency, with an approximate slope of +20 dB per decade until limited by circuit elements and op-amp bandwidth. Note that the integrator and differentiator were not performed in the experiment, but were still briefly discussed in lab.

3 Results and Analysis

3.1 Part 1: Slew Rate

The slew rate was determined from the slope of the output waveform during a square-wave transition. From the oscilloscope measurement in Figure 1:

$$\Delta V \approx 5.76 \text{ V}, \quad \Delta t \approx 6.12 \mu\text{s}. \quad (26)$$

This gives a slew rate of

$$SR \approx \frac{\Delta V}{\Delta t} \approx 0.94 \text{ V}/\mu\text{s}. \quad (27)$$

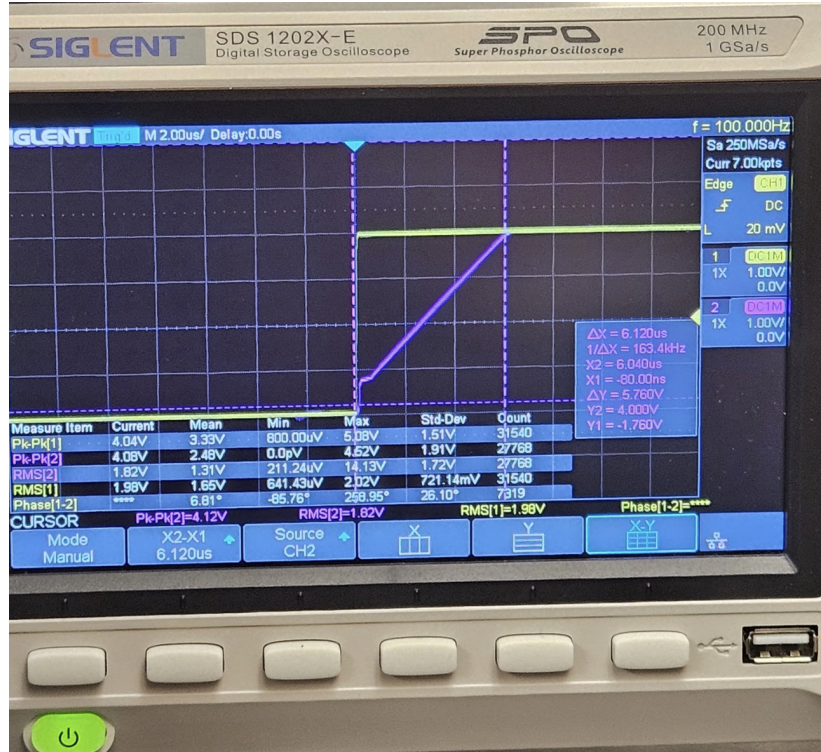


Figure 1: Measured output waveform used to determine the slew rate of the op-amp.

3.2 Part 1: Voltage Follower

For an ideal voltage follower, the output directly tracks the input, giving a gain of

$$G \approx 1, \quad (28)$$

or equivalently,

$$G \approx 0 \text{ dB.} \quad (29)$$

The measured gain remained very close to unity over the low and mid-frequency range. The maximum measured gain was

$$G_{\max} \approx 1.010, \quad (30)$$

which corresponds to

$$G_{\max, dB} \approx 0.089 \text{ dB.} \quad (31)$$

This behavior is consistent with the expected operation of a voltage follower.

At higher frequencies, the gain begins to decrease due to the finite bandwidth of the op-amp. The cut-off frequency is defined as the point where the gain decreases by 3 dB. Using the measured maximum gain, this level is

$$G_{dB} = 0.089 - 3 \approx -2.911 \text{ dB.} \quad (32)$$

From the Bode magnitude plot, the gain crosses this level between 80 kHz and 100 kHz. Interpolating between these points gives an estimated cutoff frequency of

$$f_c \approx 9.3 \times 10^4 \text{ Hz.} \quad (33)$$

Beyond this frequency, the gain decreases quickly with increasing frequency. This high frequency rolloff is expected, as the open-loop gain of the op-amp decreases with frequency.

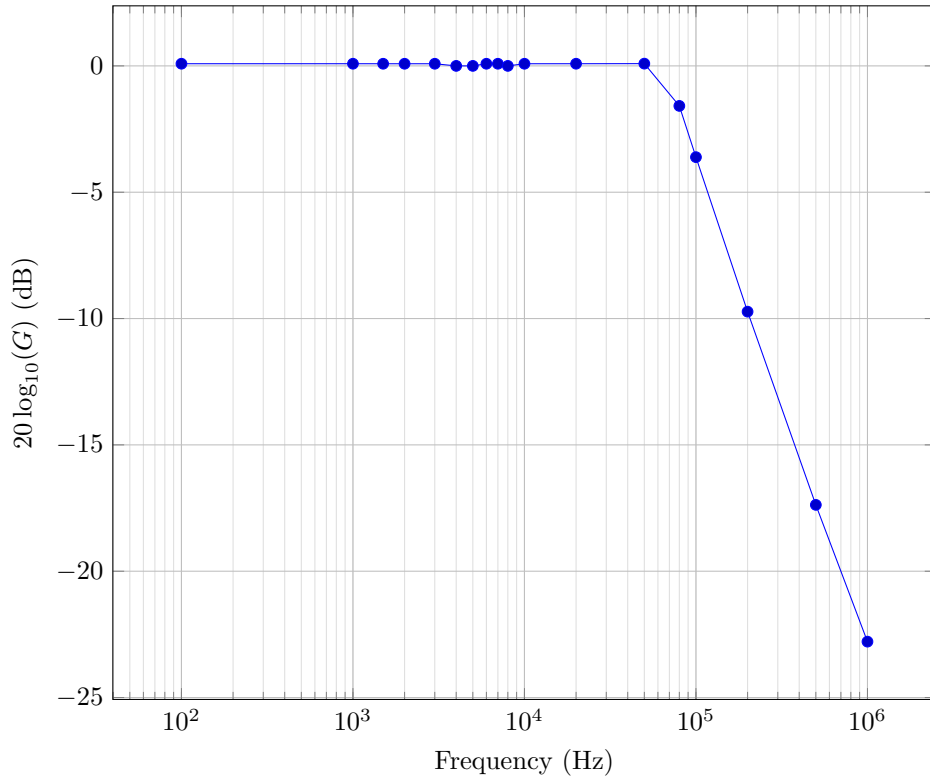


Figure 2: Bode magnitude plot for the voltage follower.

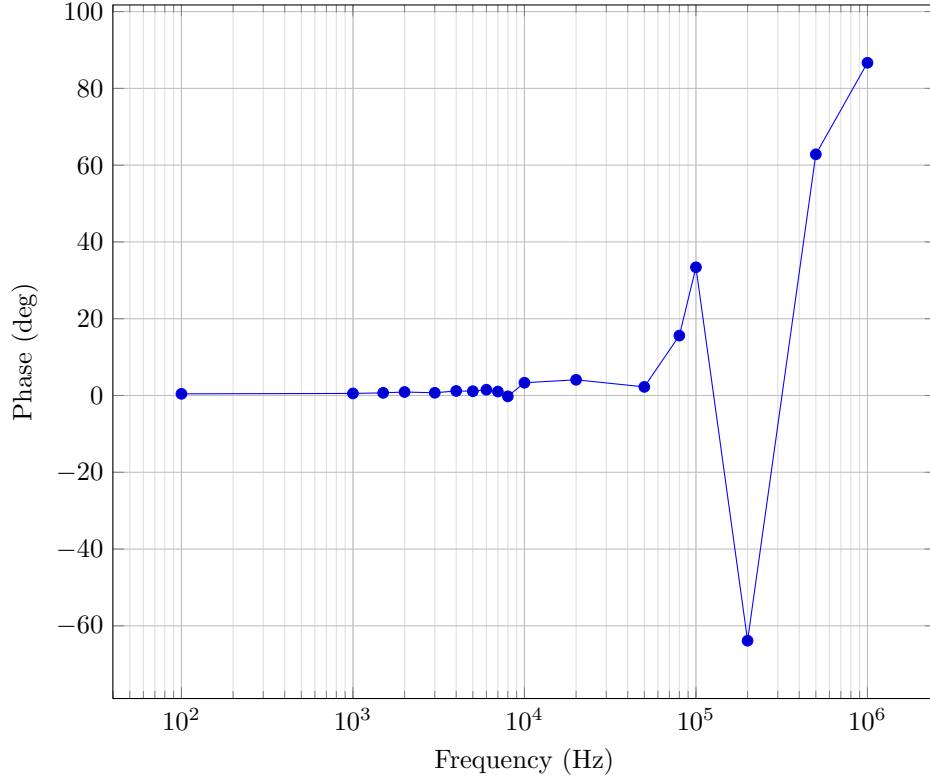


Figure 3: Phase response of the voltage follower.

3.3 Part 1: Inverting Amplifier

The inverting amplifier was constructed using $R_{in} = 993 \Omega$ and $R_f = 10.03 \text{ k}\Omega$ (experimental values). The ideal gain of the circuit is given by

$$G = -\frac{R_f}{R_{in}} \approx -10.10. \quad (34)$$

The measured gain remained approximately constant over the low- and mid-frequency range, with a typical value of

$$G \approx 9.0, \quad (35)$$

$$G_{dB} \approx 19.1 \text{ dB}. \quad (36)$$

This is slightly lower than the expected value, which can be attributed to non-ideal op-amp behavior and component tolerances.

At higher frequencies, the gain decreases due to the finite bandwidth of the op-amp. The cutoff frequency is defined as the point where the gain drops by 3 dB from its mid-band value:

$$G_{dB, cutoff} \approx 19.1 - 3 = 16.1 \text{ dB}. \quad (37)$$

From the Bode magnitude plot, the gain reaches this level between 20 kHz and 50 kHz. Interpolation gives an estimated cutoff frequency of

$$f_c \approx 3.5 \times 10^4 \text{ Hz}. \quad (38)$$

The gain-bandwidth product of the circuit can be estimated as

$$\text{GBP} = |G|f_c \approx (9.0)(3.5 \times 10^4) \approx 3.2 \times 10^5 \text{ Hz}. \quad (39)$$

At frequencies above the cutoff, the gain decreases quickly with increasing frequency. In this region the closed-loop gain becomes limited by the open-loop gain of the op-amp, and the circuit does

not maintain the ideal gain. This behavior is consistent with the expected frequency response of an inverting amplifier with finite op-amp bandwidth.

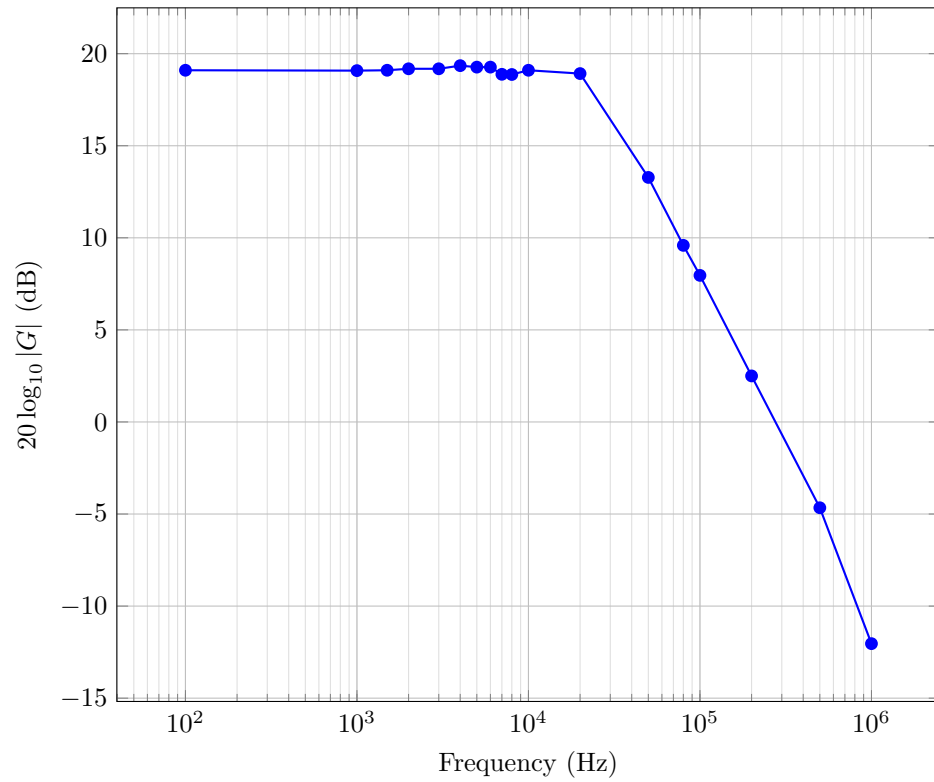


Figure 4: Bode magnitude plot for the inverting amplifier.

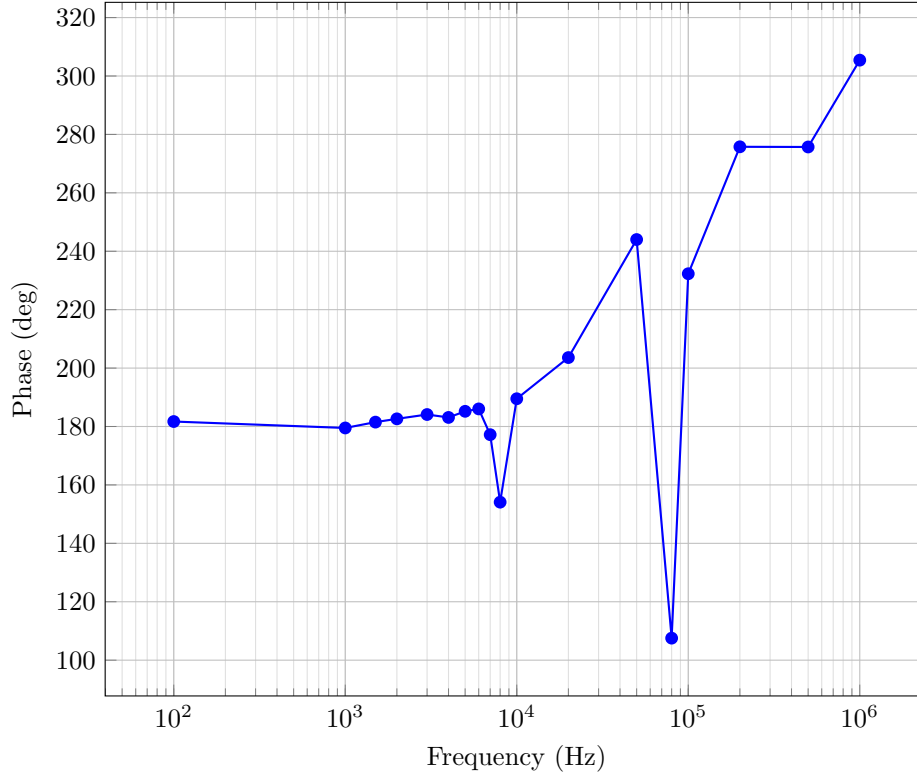


Figure 5: Phase response of the inverting amplifier.

Note that the phase values above 180° do not represent a physical phase increase. They are due to phase wrapping in the measurement.

3.4 Part 2: High-Pass Filter

The high-pass filter was constructed using a capacitor with experimental value $C = 99.6$ nF. The theoretical cutoff frequency for an RC high-pass filter is

$$f_c = \frac{1}{2\pi RC}. \quad (40)$$

The measured gain increases with frequency at low frequencies, aligning with high-pass filter behavior. At low frequencies, the gain is small, which limits the output signal. As the frequency increases, the reactance decreases and the gain rises.

The maximum measured gain was approximately

$$G_{\max} \approx 6.8, \quad (41)$$

which corresponds to

$$G_{\max, dB} \approx 16.7 \text{ dB}. \quad (42)$$

The cutoff frequency is defined as the point where the gain is 3 dB below the maximum value:

$$G_{dB, \text{cutoff}} \approx 16.7 - 3 = 13.7 \text{ dB}. \quad (43)$$

From the Bode magnitude plot, this level occurs near $f \approx 800\text{--}1000$ Hz, giving an estimated cutoff frequency of

$$f_c \approx 1.0 \times 10^3 \text{ Hz}. \quad (44)$$

Below the cutoff frequency, the gain increases approximately linearly on a logarithmic scale, corresponding to a slope of about +20 dB/decade. Above the cutoff frequency, the gain approaches a nearly constant value before decreasing at very high frequencies due to op-amp bandwidth limitations.

The phase shift also changes with frequency, transitioning from small values at low frequency toward larger phase shifts as frequency increases. This behavior is consistent with the expected response of a high-pass filter.

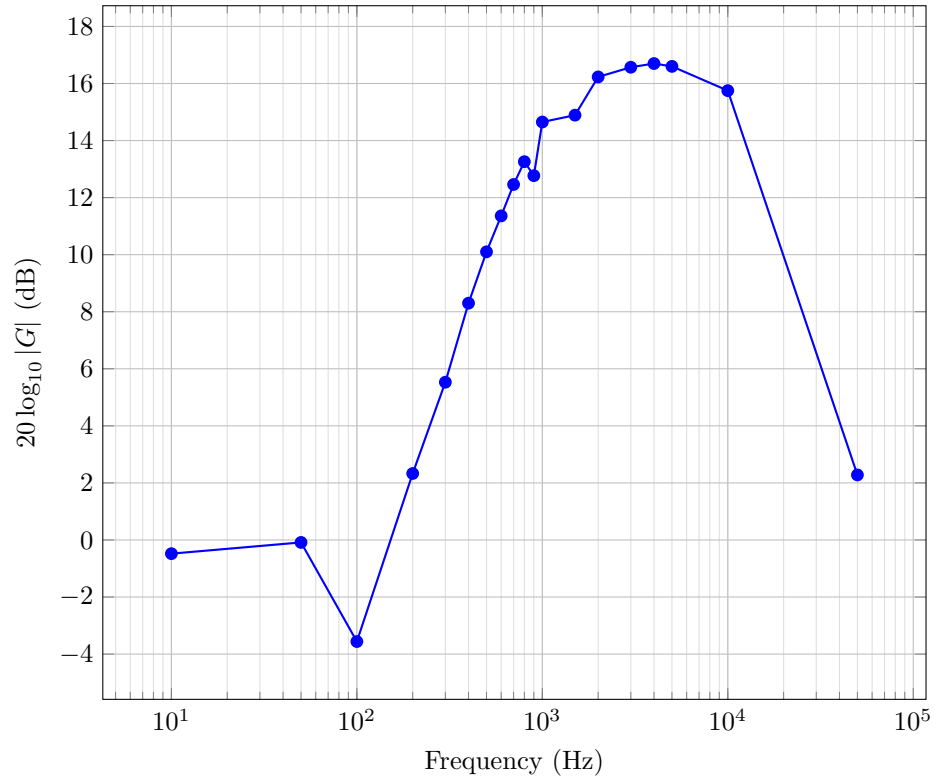


Figure 6: Bode magnitude plot for the high-pass filter.

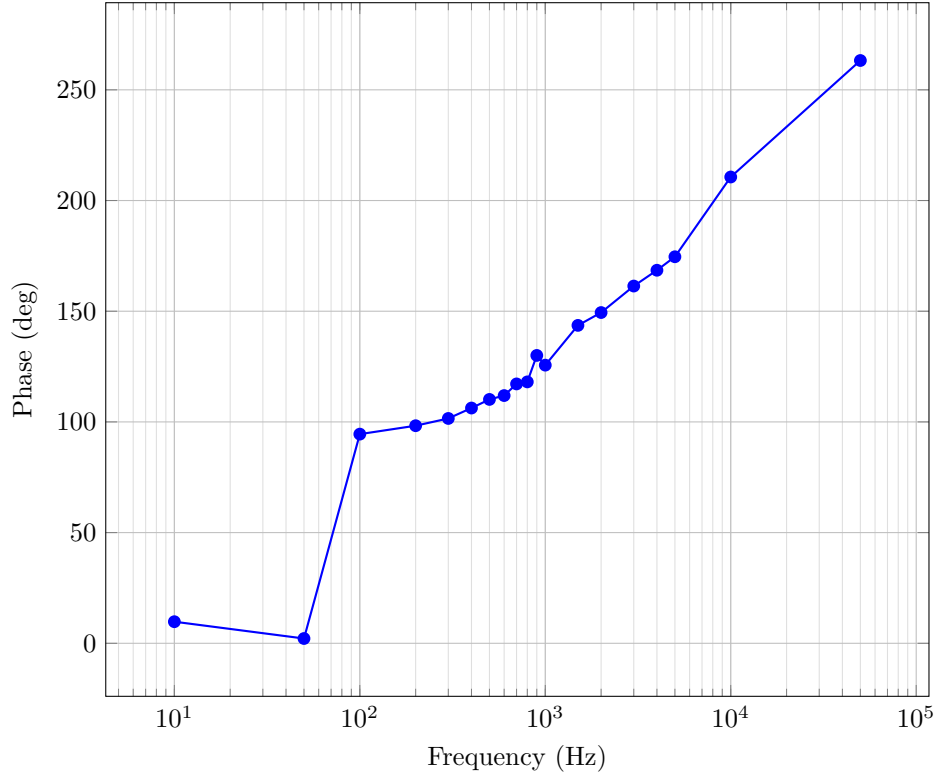


Figure 7: Phase response of the high-pass filter.

3.5 Part 2: Low-Pass Filter

The low-pass filter was constructed using an RC circuit with the output taken across the capacitor. For a low-pass filter, the cutoff frequency is given by

$$f_c = \frac{1}{2\pi RC}. \quad (45)$$

The measured gain remained roughly constant at low frequencies, which is expected for a low-pass filter. The maximum measured gain was

$$G_{\max} \approx 10.25, \quad (46)$$

which corresponds to

$$G_{\max, dB} \approx 20.2 \text{ dB}. \quad (47)$$

From the Bode magnitude plot, the gain crosses the -3 dB level between 100 Hz and 130 Hz. Interpolation gives an estimated cutoff frequency of

$$f_c \approx 1.2 \times 10^2 \text{ Hz}. \quad (48)$$

Above the cutoff frequency, the gain decreases with increasing frequency, with an approximate slope of -20 dB/decade . This behavior is consistent with the expected response of a first-order low-pass filter.

The phase shift also varies with frequency, increasing as the frequency increases. This behavior is consistent with the frequency-dependent response of the RC circuit and the op-amp circuit.

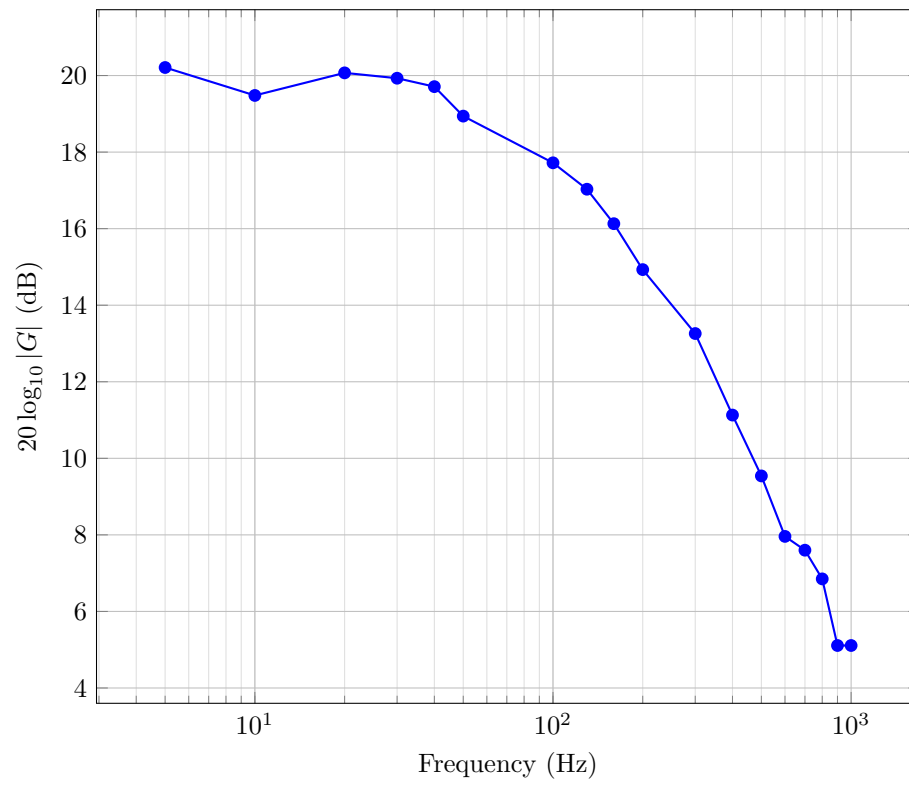


Figure 8: Bode magnitude plot for the low-pass filter.

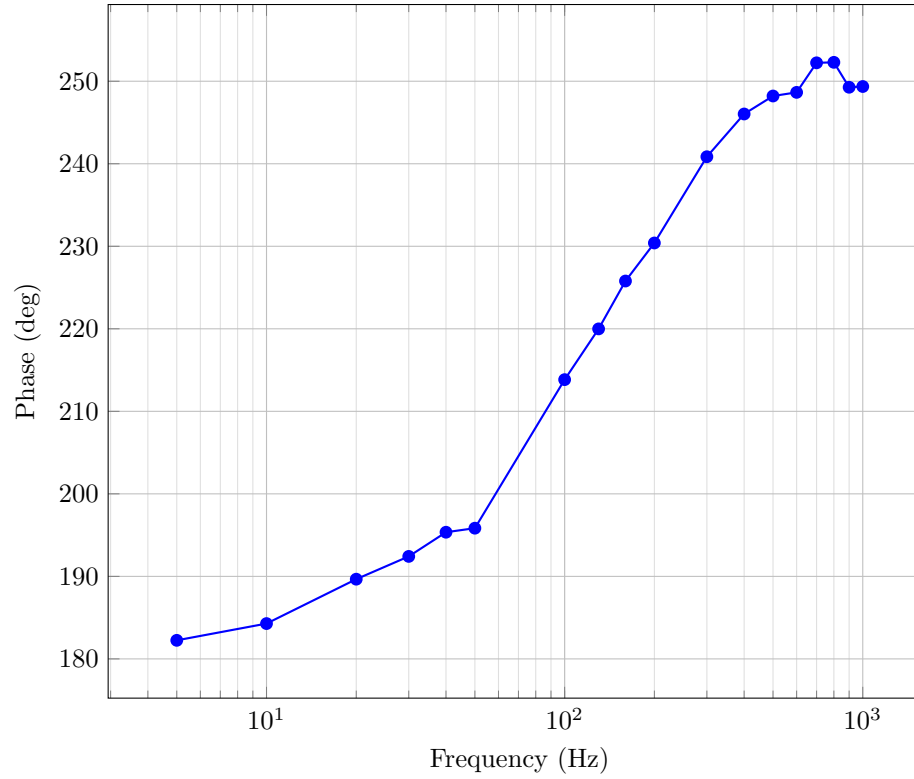


Figure 9: Phase response of the low-pass filter.

4 Conclusions

In these experiments, the behavior of operational amplifier circuits was studied using both resistive and reactive elements. For the voltage follower, the gain remained close to unity with a maximum measured value of $G_{\max} \approx 1.01$ (≈ 0.089 dB), and a cutoff frequency of approximately $f_c \approx 9.3 \times 10^4$ Hz. The inverting amplifier produced a gain close to the expected value, with a measured maximum of $G_{\max} \approx 9.28$ (≈ 19.35 dB) compared to the theoretical value of -10.10 . The cutoff frequency for this circuit was found to be $f_c \approx 3.0 \times 10^4$ Hz, demonstrating the reduction in bandwidth with increased gain. The gain-bandwidth product was approximately 3.2×10^5 Hz, consistent with the expected behavior of the op-amp.

In Part 2, the introduction of reactive elements produced frequency-dependent behavior. The high-pass filter exhibited a cutoff frequency near $f_c \approx 1.0 \times 10^3$ Hz, with gain increasing below this frequency and approaching a maximum of approximately 16.7 dB. The low-pass filter showed a cutoff frequency near $f_c \approx 1.2 \times 10^2$ Hz, with a maximum gain of approximately 20.2 dB and a rolloff consistent with -20 dB/decade at higher frequencies.

Overall, the results demonstrate that while op-amps approximate ideal behavior at low frequencies, their performance is limited by finite open-loop gain, bandwidth, and slew rate. The measured results are in good agreement with theoretical values, confirming that feedback and reactive components can be used to control gain and frequency response in circuits.

References

- [1] *Lab 8: Introduction to Operational Amplifiers.*
- [2] *Lab 9: Operational Amplifiers with Reactive Elements*
- [3] *Basic Electronics: An Introduction to Electronics for Science Students*, Curtis A. Meyer

A Responses to questions in the manual

1. Golden Rules of Op-Amp Operation

$$v_+ \approx v_-, \quad (49)$$

and

$$i_+ \approx i_- \approx 0. \quad (50)$$

These conditions come from the assumption of infinite open-loop gain and infinite input impedance.

2. Inverting Amplifier Node Voltages

In the inverting amplifier configuration, the non-inverting input is grounded so:

$$v_+ = 0. \quad (51)$$

By the golden rules,

$$v_- \approx 0, \quad (52)$$

and this is referred to as virtual ground.

The voltage drop across the input resistor is

$$V_{R_{in}} = v_{in} - v_- \approx v_{in}, \quad (53)$$

and the voltage drop across the feedback resistor is

$$V_{R_f} = v_- - v_o \approx -v_o. \quad (54)$$